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An efficient finite element method for the two-dimensional nonlinear time-space fractional Schrödinger equation on an irregular convex domain

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Abstract

In this paper, we introduce an efficient Galerkin finite element method for solving the two-dimensional nonlinear time-space fractional Schrödinger equation, with time-dependent potential function, defined on an irregular convex domain. The Caputo time fractional derivative is discretized by the finite difference scheme, while the space fractional derivative defined on the irregular convex domain is deal with the unstructured mesh finite element method. Given the nonlinear property of the considered equation, an efficient linearized strategy is used. To testify the efficiency of the proposed numerical scheme, two numerical examples are conducted with error and convergence analysis.

Keywords: two-dimensional nonlinear fractional Schrödinger equation, irregular domain, finite element method, error analysis

1. Introduction

The Schrödinger equation has played an important role in quantum mechanics. Over the past decades, with the development of the fractional calculus, fractional Schrödinger equation has been demonstrated to be able to describe more extensive fractal phenomena in quantum physics [1]. The space fractional Schrödinger equation was first generalized by Laskin [2] by extending the Brownian path integral to Lévy one. The time fractional Schrödinger equation was first considered by Naber [3] by introducing the Caputo fractional derivative instead of the first-order derivative over time to the standard Schrödinger equation.

In recent years, research on the nonlinear fractional Schrödinger equation has attracted wide attention [4, 5]. For example, Zhao et al. [6] deduced a fourth-order compact ADI scheme for a two-dimensional nonlinear space fractional Schrödinger equation. Li et al. [7] considered a class of nonlinear one-dimensional Riesz space-fractional Schrödinger equations using Galerkin finite element method. Zhang et al. [8] proposed a Crank-Nicolson Galerkin-Legendre spectral scheme for the one-dimensional and two-dimensional nonlinear space fractional Schrödinger equation. Li et al.[9] proposed a linearized L1-Galerkin finite element method to solve the multi-dimensional nonlinear time-fractional Schrödinger equation.

In the most existing literatures, research on the time-space fractional Schrödinger equations on two dimensions is still limited. Further, attention is mainly focused on the equations defined on regular domains. Given that most of the research subjects in practical problem tend to be irregular, research on the numerical methods for solving the two-dimensional nonlinear time-space fractional Schrödinger equations on an irregular convex domain is still an important issue.

In this paper, we consider the following two-dimensional nonlinear time-space fractional Schrödinger equation (2D-NFSE) on an irregular convex domain:

$$i\frac{\partial^\alpha \Psi}{\partial t^\alpha} + \kappa\left(\frac{\partial^{2\beta} \Psi}{\partial |x|^{2\beta}} + \frac{\partial^{2\beta} \Psi}{\partial |y|^{2\beta}}\right) + v(x, y, t)\Psi + \lambda|\Psi|^2\Psi = f(x, y, t), \quad (x, y, t) \in \Omega \times [0, T], \quad (1.1)$$

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subject to the initial condition

$$\Psi(x, y, 0) = \phi(x, y), (x, y) \in \Omega, \quad (1.2)$$

and zero Dirichlet boundary condition

$$\Psi(x, y, t) = 0, (x, y) \in \partial\Omega, \quad (1.3)$$

where $0 < \alpha < 1$, $1/2 < \beta < 1$, κ, λ are constants and $v(x, y, t)$ is a time-dependent potential function. The irregular convex domain Ω is defined as $\Omega = \{(x, y) | a(y) \leq x \leq b(y), c(x) \leq y \leq d(x)\}$, where $a(y), b(y)$ are the left and right boundaries of Ω , and $c(x), d(x)$ are the lower and upper boundaries of Ω . $\frac{\partial^\alpha \Psi}{\partial t^\alpha}$ is the Caputo fractional derivative [10].

For a function $\Psi(x, y, t)$, the Riesz space fractional derivatives $\frac{\partial^{2\beta} \Psi}{\partial |x|^{2\beta}}$ and $\frac{\partial^{2\beta} \Psi}{\partial |y|^{2\beta}}$ are defined in [11] by

$$\frac{\partial^{2\beta} \Psi(x, y, t)}{\partial |x|^{2\beta}} = -c_\beta \left({}_x \hat{D}_L^{2\beta} \Psi(x, y, t) + {}_x \hat{D}_R^{2\beta} \Psi(x, y, t) \right), \frac{\partial^{2\beta} \Psi(x, y, t)}{\partial |y|^{2\beta}} = -c_\beta \left({}_y \hat{D}_L^{2\beta} \Psi(x, y, t) + {}_y \hat{D}_R^{2\beta} \Psi(x, y, t) \right), \quad (1.4)$$

where $c_\beta = \frac{1}{2 \cos(\beta\pi)}$, and the Riemann-Liouville fractional derivative operators [12] with $1/2 < \beta < 1$ are defined as

$${}_x \hat{D}_L^{2\beta} \Psi(x, y, t) = \frac{1}{\Gamma(2-2\beta)} \frac{\partial^2}{\partial x^2} \int_{a(y)}^x (x-s)^{1-2\beta} \Psi(s, y, t) ds, {}_x \hat{D}_R^{2\beta} \Psi(x, y, t) = \frac{1}{\Gamma(2-2\beta)} \frac{\partial^2}{\partial x^2} \int_x^{b(y)} (s-x)^{1-2\beta} \Psi(s, y, t) ds. \quad (1.5)$$

With respect to y , the fractional derivative operators ${}_y \hat{D}_L^{2\beta} \Psi(x, y, t)$, ${}_y \hat{D}_R^{2\beta} \Psi(x, y, t)$ are defined similarly.

2. Numerical method

2.1. Some preliminaries

In this section, we first recall some definitions and lemmas that needed in the finite element numerical analysis.

Definition 1. ([13, 14, 15]) As for a convex domain $\Omega \subset \mathbb{R}^2$, resulting from its irregularity, with $x_{\min} = \min_{(x,y) \in \Omega} a(y)$, $x_{\max} = \max_{(x,y) \in \Omega} b(y)$, $y_{\min} = \min_{(x,y) \in \Omega} c(x)$ and $y_{\max} = \max_{(x,y) \in \Omega} d(x)$, we define the inner product and L^2 -norm as

$$(u, v)_{L^2(\Omega)} := \int_{\Omega} u \bar{v} d\Omega = \int_{y_{\min}}^{y_{\max}} \int_{a(y)}^{b(y)} u(x, y) \bar{v}(x, y) dx dy = \int_{x_{\min}}^{x_{\max}} \int_{c(x)}^{d(x)} u(x, y) \bar{v}(x, y) dy dx, \quad (2.6)$$

$$\|u\|_{L^2(\Omega)} = ((u, u)_{L^2(\Omega)})^{1/2}. \quad (2.7)$$

Lemma 1. ([15]) If $\mu \in (1, 2)$, $u, v \in J_{L,0}^\mu(\Omega)$ (or $J_{R,0}^\mu(\Omega)$) (defined in Definition 2), then

$$({}_x \hat{D}_L^\mu u, v)_{L^2(\Omega)} = ({}_x \hat{D}_L^{\mu/2} u, {}_x \hat{D}_R^{\mu/2} v)_{L^2(\Omega)}, ({}_y \hat{D}_L^\mu u, v)_{L^2(\Omega)} = ({}_y \hat{D}_L^{\mu/2} u, {}_y \hat{D}_R^{\mu/2} v)_{L^2(\Omega)},$$

$$({}_x \hat{D}_R^\mu u, v)_{L^2(\Omega)} = ({}_x \hat{D}_R^{\mu/2} u, {}_x \hat{D}_L^{\mu/2} v)_{L^2(\Omega)}, ({}_y \hat{D}_R^\mu u, v)_{L^2(\Omega)} = ({}_y \hat{D}_R^{\mu/2} u, {}_y \hat{D}_L^{\mu/2} v)_{L^2(\Omega)}.$$

Definition 2. ([13, 14]) (Left fractional derivative space). For $\mu > 0$, we define the semi-norm and norm separately as $|u|_{J_L^\mu(\Omega)} := \left(\|{}_x \hat{D}_L^\mu u\|_{L^2(\Omega)}^2 + \|{}_y \hat{D}_L^\mu u\|_{L^2(\Omega)}^2 \right)^{\frac{1}{2}}$, $\|u\|_{J_L^\mu(\Omega)} := \left(\|u\|_{L^2(\Omega)}^2 + |u|_{J_L^\mu(\Omega)}^2 \right)^{\frac{1}{2}}$, where $J_L^\mu(\Omega)$, $J_{L,0}^\mu(\Omega)$ denote the closure of $C^\infty(\Omega)$, $C_0^\infty(\Omega)$ with respect to $\|\cdot\|_{J_L^\mu(\Omega)}$.

The Right fractional derivative space $(|u|_{J_R^\mu(\Omega)}, \|u\|_{J_R^\mu(\Omega)})$, the Fractional Sobolev space $(|u|_{H^\mu(\Omega)}, \|u\|_{H^\mu(\Omega)})$ and the Symmetric fractional derivative space $(|u|_{J_s^\mu(\Omega)}, \|u\|_{J_s^\mu(\Omega)})$ can also be defined similarly [15, 16]. Throughout the proceeding sections, we denote $(\cdot, \cdot) = (\cdot, \cdot)_{L^2(\Omega)}$, $\|\cdot\|_0 = \|\cdot\|_{L^2(\Omega)}$.

2.2. Numerical method

Let $\tau = T/N$ be the time step, $t_n = n\tau$, $n = 0, 1, \dots, N$. Denote $\Psi(x, y, t_n) = \Psi^n$ as the exact solution at time $t = t_n$, and Ψ_h^n is the corresponding numerical solution. To numerically solve the 2D-NFSE defined in the Eq. (1.1), we first discretize the Caputo time fractional derivative by the finite difference scheme [17].

$${}_0^C D_t^\alpha \Psi^n = \frac{\tau^{-\alpha}}{\Gamma(2-\alpha)} \sum_{j=0}^{n-1} a_j^\alpha (\Psi^{n-j} - \Psi^{n-j-1}) + R_n = \frac{\tau^{-\alpha}}{\Gamma(2-\alpha)} \left[\Psi^n - \sum_{j=1}^{n-1} (a_{n-j-1}^\alpha - a_{n-j}^\alpha) \Psi^j - a_{n-1}^\alpha \Psi^0 \right] + R_n, \quad (2.8)$$

the coefficients $a_j^\alpha = (j+1)^{1-\alpha} - j^{1-\alpha}$, $j = 0, 1, \dots, n-1$ satisfy $a_0^\alpha = 1$, $\sum_{j=1}^{n-1} (a_{n-j-1}^\alpha - a_{n-j}^\alpha) + a_{n-1}^\alpha = 1$. The truncation error $|R_n| \leq C \max_{0 \leq t \leq T} |\frac{\partial^2 \Psi}{\partial t^2}| \tau^{2-\alpha}$, where C is a positive constant.

For $n \geq 1$, we denote

$$\nabla_t^\alpha \Psi^n = \frac{\tau^{-\alpha}}{\Gamma(2-\alpha)} \left[\Psi^n - \sum_{j=1}^{n-1} (a_{n-j-1}^\alpha - a_{n-j}^\alpha) \Psi^j - a_{n-1}^\alpha \Psi^0 \right]. \quad (2.9)$$

Assume that $\{\mathcal{A}_h\}$ is a family of unstructured triangulations of the convex domain Ω and h is the maximum diameter of the triangular elements in $\mathcal{A}_h$. The finite dimensional subspace $X_h \in X$, where $X = H_0^\beta(\Omega)$, is defined as $X_h = \{\chi_h | \chi_h \in C(\Omega), \chi_h \in X, \chi_h|_K \in P_r(K), \forall K \in \mathcal{A}_h\}$, where $P_r(K)$ is the set of polynomials with degree at most r in triangular element K . Then we can get the fully discrete scheme of the considered problem: to find $\Psi_h^n \in X_h$ for $n = 1, 2, \dots, N$ satisfies

$$\begin{cases} i(\nabla_t^\alpha \Psi_h^n, \chi_h) - \Lambda(\Psi_h^n, \chi_h) + (v^n \Psi_h^n, \chi_h) + \lambda(|\Psi_h^n|^2 \Psi_h^n, \chi_h) = (f^n, \chi_h), \forall \chi_h \in X_h, \\ \Psi_h^0 = P_h \phi(x, y), \end{cases} \quad (2.10)$$

where $P_h : X \rightarrow X_h$ is a projection operator. The bilinear form $\Lambda(\Psi, \chi)$ is derived based on the Lemma 1 as

$$\Lambda(\Psi, \chi) = \kappa c_\beta \{ ({}_x \hat{D}_L^\beta \Psi, {}_x \hat{D}_R^\beta \chi) + ({}_x \hat{D}_R^\beta \Psi, {}_x \hat{D}_L^\beta \chi) \} + \kappa c_\beta \{ ({}_y \hat{D}_L^\beta \Psi, {}_y \hat{D}_R^\beta \chi) + ({}_y \hat{D}_R^\beta \Psi, {}_y \hat{D}_L^\beta \chi) \}. \quad (2.11)$$

Considering the nonlinear property, to cut the cost as much as possible, by Taylor's theorem, we employ a feasible linearized strategy to treat the nonlinear part [9], that is, when $n \geq 2$,

$$|\Psi_h^n|^2 \Psi_h^n = |\hat{\Psi}_h^n|^2 \Psi_h^n + O(\tau^2), \quad \hat{\Psi}_h^n = 2\Psi_h^{n-1} - \Psi_h^{n-2}. \quad (2.12)$$

Denote $\mu = \frac{\tau^{-\alpha}}{\Gamma(2-\alpha)}$, then we have for $n \geq 2$,

$$i\mu(\Psi_h^n, \chi_h) - \Lambda(\Psi_h^n, \chi_h) + (v^n \Psi_h^n, \chi_h) + \lambda(|\hat{\Psi}_h^n|^2 \Psi_h^n, \chi_h) = i\mu \sum_{j=1}^{n-1} (a_{n-j-1}^\alpha - a_{n-j}^\alpha) (\Psi_h^j, \chi_h) + i\mu a_{n-1}^\alpha (\Psi_h^0, \chi_h) + (f^n, \chi_h). \quad (2.13)$$

When $n = 1$, we use the following approximation

$$i\mu(\Psi_h^1, \chi_h) - \Lambda(\Psi_h^1, \chi_h) + (v^1 \Psi_h^1, \chi_h) + \lambda(|\Psi_h^0|^2 \Psi_h^1, \chi_h) = i\mu(\Psi_h^0, \chi_h) + (f^1, \chi_h). \quad (2.14)$$

For the convex domain considered in this paper, we utilize the unstructured triangulation, which has been demonstrated to be much more flexible and can lead to a better performance with smaller errors than the commonly used structured mesh [15], to deal with the domain Ω . Define $\{(x_l, y_l) : l = 1, 2, \dots, \mathcal{N}\}$ as the nodes, where $\mathcal{N}$ is the total number of the nodes in the mesh. Let $\varphi_k(x_l, y_l) = \delta_{kl}$, ($k, l = 1, 2, \dots, \mathcal{N}$) be the basis functions, where δ_{kl} is the Kronecker delta symbol. Then, for each time step $t = t_n$, the finite element solution Ψ_h^n can be expressed as

$$\Psi_h^n = \sum_{k=1}^{\mathcal{N}} \Psi_k^n \varphi_k(x, y). \quad (2.15)$$

Choosing $\chi_h = \varphi_l(x, y) (l = 1, 2, \dots, N)$, then the Eqs.(2.13)-(2.14) can be rewritten as the following matrix forms:

$$(i\mu \mathbf{M} - \mathbf{A} + \widetilde{\mathbf{M}}^n + \lambda \widetilde{\mathbf{W}}^{n-1}) \Psi^n = i\mu \sum_{j=1}^{n-1} (a_{n-j-1}^\alpha - a_{n-j}^\alpha) \mathbf{M} \Psi^j + i\mu a_{n-1}^\alpha \mathbf{W}^0 + \mathbf{F}^n, \quad (n \geq 2) \quad (2.16)$$

$$(i\mu \mathbf{M} - \mathbf{A} + \widetilde{\mathbf{M}}^1 + \lambda \widetilde{\mathbf{W}}^0) \Psi^1 = i\mu \mathbf{W}^0 + \mathbf{F}^1, \quad (n = 1) \quad (2.17)$$

where $\Psi^n = (\Psi_1^n, \Psi_2^n, \dots, \Psi_N^n)^T$ is the unknown numerical solution vector. The matrix $\mathbf{M} = ((\varphi_k, \varphi_l))_{N \times N}$ is the mass matrix, and $\mathbf{A} = (\Lambda(\varphi_k, \varphi_l))_{N \times N}$ is the stiffness matrix. We also have the matrixes $\widetilde{\mathbf{M}}^n = ((v^n \varphi_k, \varphi_l))_{N \times N}$, $\widetilde{\mathbf{W}}^0 = ((|\Psi_h^0|^2 \varphi_k, \varphi_l))_{N \times N}$, $\widetilde{\mathbf{W}}^{n-1} = ((|\Psi_h^{n-1} - \Psi_h^{n-2}|^2 \varphi_k, \varphi_l))_{N \times N}$, $(n \geq 2)$, $\mathbf{W}^0 = ((\Psi_h^0, \varphi_l))_{N \times 1}$ and $\mathbf{F}^n = ((f^n, \varphi_l))_{N \times 1}$.

3. Numerical examples

To testify the efficiency of the proposed finite element method for the 2D-NFSE defined on an irregular domain, we give two numerical examples in this section. At time $t_N = T$, the exact solution can be denoted by its real and imaginary part as $\Psi(x, y, T) = u(x, y, T) + iz(x, y, T)$, the corresponding numerical solution is denoted as $\Psi_h^N = U^N + iZ^N$. The error norm can be expressed as

$$Err = \|\Psi(x, y, T) - \Psi_h^N\|_0 = (\|u(x, y, T) - U^N\|_0^2 + \|z(x, y, T) - Z^N\|_0^2)^{1/2} = (Err_1^2 + Err_2^2)^{1/2}, \quad (3.18)$$

where $Err_1 = (\|u(x, y, T) - U^N\|_0^2)^{1/2}$ and $Err_2 = (\|z(x, y, T) - Z^N\|_0^2)^{1/2}$ indicating the errors in real part and in imaginary part, separately. The convergence order can be calculated by the following formulation:

$$Order = \begin{cases} \frac{\log(error(h_1)/error(h_2))}{\log(h_1/h_2)}, & \text{in space,} \\ \frac{\log(error(\tau_1)/error(\tau_2))}{\log(\tau_1/\tau_2)}, & \text{in time,} \end{cases}$$

where $error(h_1)$ denotes the errors Err or Err_1 or Err_2 with space step being h_1 .

3.1. Example 1

In this example, we consider the following nonlinear problem on an ellips convex region $\Omega = \{(x, y) | \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1\}$:

$$\begin{cases} i \frac{\partial^\alpha \Psi}{\partial t^\alpha} + \kappa (\frac{\partial^{2\beta} \Psi}{\partial |x|^{2\beta}} + \frac{\partial^{2\beta} \Psi}{\partial |y|^{2\beta}}) + v(x, y, t) \Psi(x, y, t) + \lambda |\Psi|^2 \Psi = f(x, y, t), & (x, y, t) \in \Omega \times [0, T], \\ \Psi(x, y, 0) = \varphi(x, y), & (x, y) \in \Omega, \\ \Psi(x, y, t) = 0, & (x, y) \in \partial\Omega, \end{cases} \quad (3.19)$$

with $\lambda = \kappa = 1$, $v(x, y, t) = t(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1)$, $\varphi(x, y) = i(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1)^2$,

$$\begin{aligned} f(x, y, t) = & [i \frac{t^{1-\alpha}}{\Gamma(2-\alpha)} - \frac{2t^{2-\alpha}}{\Gamma(3-\alpha)}] (\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1)^2 - \frac{t + i(t^2 + 1)}{2 \cos(\beta\pi)} [q(x, a) + q(y, b)] \\ & + [t + i(t^2 + 1)] t (\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1)^3 + [t^2 + (t^2 + 1)^2] [t + i(t^2 + 1)] (\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1)^6. \end{aligned} \quad (3.20)$$

With $s_1(a) = \frac{\Gamma(5)}{a^4 \Gamma(5-2\beta)}$, $s_2(a) = \frac{\Gamma(4)}{a^4 \Gamma(4-2\beta)}$, $s_3(a) = \frac{\Gamma(3)}{a^4 \Gamma(3-2\beta)}$, the function $q(x, a)$ ($q(y, b)$) can be given as

$$q(x, a) = s_1(a) [(x-x_l)^{4-2\beta} + (x_r-x)^{4-2\beta}] + 4s_2(a) [x_l(x-x_l)^{3-2\beta} - x_r(x_r-x)^{3-2\beta}] + 4s_3(a) x_l^2 (x-x_l)^{2-2\beta} + 4s_3(a) x_r^2 (x_r-x)^{2-2\beta}, \quad (3.21)$$

where $x_l = -\frac{a}{b} \sqrt{b^2 - y^2}$, $x_r = \frac{a}{b} \sqrt{b^2 - y^2}$, $y_l = -\frac{b}{a} \sqrt{a^2 - x^2}$, $y_r = \frac{b}{a} \sqrt{a^2 - x^2}$.

The exact solution is $\Psi(x, y, t) = [t + i(t^2 + 1)] (\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1)^2$. Taking $h = 1/48$, $\tau = 1/1000$, $T = 1$, $a = 1/2$, $b = 1/4$, Fig.1 shows the comparison of the exact solution and the numerical solution of the considered nonlinear fractional Schrödinger equation. From the figures, the numerical solution is well consistent with the exact solution. It is demonstrated that the numerical scheme is efficient and correct.

The Table 1 shows the numerical errors Err , as well as the errors in real part Err_1 and that in imaginary part Err_2 , with the arbitrary fractional derivative orders. With the time step fixed as $\tau = 1/1000$, the convergence orders in space are obtained to be 2. It is consistent with the finite element theoretical in space [15], indicating that the numerical scheme is valid.

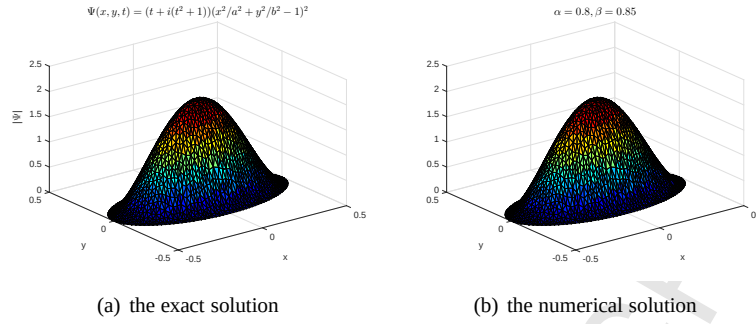


Figure 1: The numerical solution compared with the exact solution of the considered 2D-NFSE.

Table 1: The convergence order when $\alpha = 0.8, \beta = 0.85$ with $\tau = 1/1000$.

	h	Err	order	Err ₁	order	Err ₂	order
$\alpha = 0.8$ $\beta = 0.85$	1/8	5.20734E-02	-	2.29469E-02	-	4.67448E-02	-
	1/16	1.29953E-02	2.0026	5.76524E-03	1.9928	1.16465E-02	2.0049
	1/32	3.22623E-03	2.0101	1.43517E-03	2.0062	2.88943E-03	2.0071
	1/48	1.41032E-03	2.0409	6.26875E-04	2.0428	1.26334E-03	2.0404

3.2. Example 2

In this example, we consider the following problem on a rectangular region $\Omega = [0, 1] \times [0, 1]$:

$$\begin{cases} i \frac{\partial^\alpha \Psi}{\partial t^\alpha} + \kappa \left(\frac{\partial^{2\beta} \Psi}{\partial |x|^{2\beta}} + \frac{\partial^{2\beta} \Psi}{\partial |y|^{2\beta}} \right) + v(x, y, t) \Psi(x, y, t) + \lambda |\Psi|^2 \Psi = f(x, y, t), & (x, y, t) \in \Omega \times [0, T], \\ \Psi(x, y, 0) = \varphi(x, y), & (x, y) \in \Omega, \\ \Psi(x, y, t) = 0, & (x, y) \in \partial\Omega, \end{cases} \quad (3.22)$$

with $\lambda = \kappa = 1$, $v(x, y, t) = tx(1-x)y(1-y)$, $\varphi(x, y) = 15ix^2(1-x)^2y^2(1-y)^2$,

$$\begin{aligned} f(x, y, t) = & i \left[\frac{15t^{1-\alpha}}{\Gamma(2-\alpha)} - \frac{30t^{1-\alpha}}{\Gamma(2-\alpha)} - \frac{30t^{2-\alpha}}{\Gamma(3-\alpha)} \right] x^2(1-x)^2y^2(1-y)^2 \\ & + 3375[t^2 + (1+t)^4][t + i(1+t)^2]x^6(1-x)^6y^6(1-y)^6 + 15[t + i(1+t)^2]tx^3(1-x)^3y^3(1-y)^3 \\ & - \frac{15[t + i(1+t)^2]y^2(1-y)^2}{\cos(\beta\pi)} [p(x, \beta) + q(x, \beta)] - \frac{15[t + i(1+t)^2]x^2(1-x)^2}{\cos(\beta\pi)} [p(y, \beta) + q(y, \beta)]. \end{aligned} \quad (3.23)$$

$$p(x, \beta) = \frac{x^{2-2\beta}}{\Gamma(3-2\beta)} \left[1 - \frac{6x}{3-2\beta} + \frac{12x^2}{(3-2\beta)(4-2\beta)} \right], q(x, \beta) = \frac{(1-x)^{2-2\beta}}{\Gamma(3-2\beta)} \left[1 - \frac{6(1-x)}{3-2\beta} + \frac{12(1-x)^2}{(3-2\beta)(4-2\beta)} \right]. \quad (3.24)$$

The exact solution for this problem can be obtained as $\Psi(x, y, t) = 15[t + i(1+t)^2]x^2(1-x)^2y^2(1-y)^2$. Taking $h = 1/32, \tau = 1/1000, T = 1$, the numerical solution is shown in Fig.(2), which is in well accordance with the exact solution, indicating that the numerical scheme is valid and efficient. The Table 2 shows the errors and convergence analysis in space. The convergence order is also obtained to be 2. It is demonstrated that the numerical scheme is efficient.

4. Conclusions

In this paper, we propose an efficient unstructured mesh finite element method for solving the two-dimensional nonlinear time-space fractional Schrödinger equation defined on an irregular convex domain. The potential function is considered to be time-dependent. A fully discrete numerical scheme is deduced. To testify the efficiency of the proposed method, two numerical examples are given with error and convergence analysis. This paper provides an efficient numerical method for solving two-dimensional nonlinear fractional Schrödinger equation on irregular convex domains.

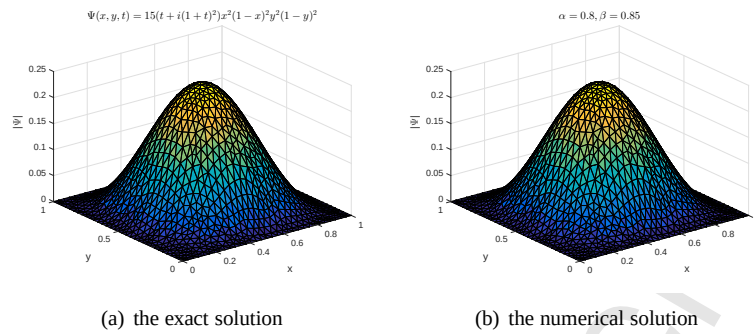


Figure 2: The numerical solutions compared with the exact solution of the considered 2D-NFSE.

Table 2: The convergence order when $\alpha = 0.8, \beta = 0.85$ with $\tau = 1/1000$.

	h	Err	order	Err ₁	order	Err ₂	order
$\alpha = 0.8$	1/8	2.83477E-03	-	6.69221E-04	-	2.75464E-03	-
	1/16	7.00525E-04	2.0167	1.65013E-04	2.0199	6.80813E-04	2.0165
$\beta = 0.85$	1/24	3.05361E-04	2.0479	7.18833E-05	2.0494	2.96779E-04	2.0478
	1/32	1.66471E-04	2.1088	3.88318E-05	2.1406	1.61878E-04	2.1070

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